# Constraining the Nature of Dark Matter Subhalos with Lensed Gravitational Waves

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Gravitational waves offer an unprecedented probe into the small scale structure of dark matter halos. Leveraging their exquisite time resolution, we show that strongly lensed gravitational waves will allow for both the strong constraint of the Self-Interacting Dark Matter cross section, and Warm Dark Matter structure formation compared to the Cold Dark Matter paradigm. In this *Letter*, we present the rate at which future detections of lensed gravitational waves in the geometric-optics regime will produce indications of the existence of dark matter sub-structure for these three models. We find that with ground-based detectors, over an observing period of 3 years, the detection of additional images generated by core-collapsed subhalos down to masses of $10^5 M_\odot$ will be able to distinguish between our fiducial cold and warm dark matter models by $\sim 2.5 - 4.7\ \sigma$, for optimistic merger rates, and by $\sim 0.7 - 1.3\ \sigma$ for conservative merger rates. Even with conservative rates, a single additional image detected during 10 years of A$^\sharp$ observations would disfavor cold dark matter at $\sim 2.2 - 2.4\ \sigma$, although such a detection is expected in only $8 - 16\%$ of realizations. Finally, we report that the existence of collapsed low mass subhalos will produce a median of $\sim 5 - 6$ times greater wave-optics effects in signals detected by future space-based detectors when compared against cold dark matter, further motivating the use of gravitational waves as a premier probe of the nature of dark matter at small scales.

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